# Optimal Digital Filtering in Gamma-ray Spectroscopy

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***Abstract*— In this paper, we address the subject of the digital nuclear spectroscopy as seen as a counterpart of the classic analogue approach. Consequently, we will present the design as well as the implementation on a DSP board, of the various necessary digital pulse processing techniques via digital filtering in order to provide the principal tasks which often take place in a generic "Gamma" digital spectroscopic setup.**

**The first part will be devoted to the design of the digital IIR filter used for the charge preamplifier's slow-pole compensation. This will be followed by the practical estimation of the power spectral density relating to the electrical noise components present at the spectrometer's input. Thereafter, a very detailed attention will be given to the design of the digital optimal filter to be used for the charge measurements. We follow by another FIR filter, this one deals with the digital estimation of the reference line of measurements and then to achieve the role of the: *Base Line Restorer*. Lastly, we give a hardware implementation of the designed filters on the board: "*TMS320C6713-DSK*", a DSP KIT developed by "*DIGITAL Spectrum*".**

***Index Terms*— Optimal filtering, DSProcessor, IIR digital filter, FIR digital filter, Gamma-ray spectroscopy, Optimization under constraints.**

## I. Introduction

The measuring equipments used in nuclear spectroscopy are particularly sensitive to the ***signal to noise ratio (SNR)***, which is a determining factor to ensure a great quality of measurement [1]. The widening of the final spectrum lines (bad resolution) denounces inaccuracies of measurement. One of the principal causes of these erroneous measurements is the electronic noise introduced by the used equipment and this especially when the useful signal is relatively small. We note also effects related to the *D.C.* floor fluctuations, pulse pileup effects, electromagnetic and power supply disturbances, thermal and electromagnetic instabilities and disturbances of the power supply, …etc [2]. In a so close past, the fact of trying to eliminate or even to compensate all these sources of errors at the same time, concerns the tasks of an utopian measuring instrument; especially when we are using traditional equipments (purely analog) which cannot meet one requirement without causing a wrong to another one. Fortunately, this situation has changed during the last few years with the introduction of the digital signal processing techniques into the field of the nuclear Gamma-ray spectroscopy. And we have assisted to the emergence of several methods, which were proposed and demonstrated in specialized literature, in particular for the « $\gamma$ » and « $X$ » ray spectroscopy [3-8].

In fact, the digital processing of the signal emanating from the nuclear detector, allows the synthesis of almost any shape of the "*weighting function*" related to the sought filters in the context of the spectroscopy of high precision. Thus we can get filters so close to those theoretically known as optimal ones [6], with an intrinsic adaptability to the various unexpected components of electrical noise in a given experiment. In a typical architecture of the digitized spectroscopic chain (see Fig. 1.), we generally samples the signal at the output of an analog pre-filter (charge preamplifier + anti-aliasing filter) at a frequency adapted to the current context of the considered experiment (generally in the range of 5MHz ~ 50Mhz). The samples are processed either with a dedicated numerical electronics or via the execution of algorithms implemented on board containing DSP and/or FPGA circuits and the resulting spectrum is often plotted on a PC's monitor.

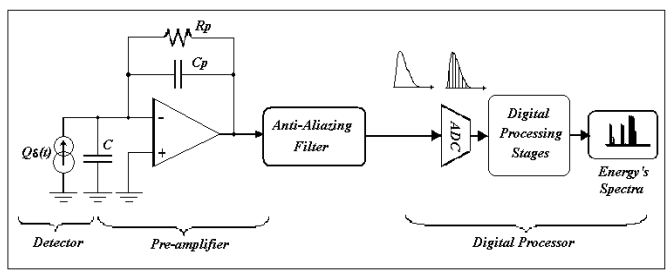


Fig. 1 Typical architecture of the digitized Gamma-ray spectroscopic chain.

The rest of the paper will be organized as follows: In the next section, we explain extensively the design of the main used digital filters. Section III describes briefly the implementation of the designed digital filters on a DSP board. Finally, conclusion is presented in section IV.

## II. Design of the Main Digital Filters

Generally, The main implemented digital filters in the context of the digital Gamma-ray spectroscopy are: i) The pole-zero digital compensator, ii) The baseline digital restorer, iii) The energy digital estimator. For more options we can add

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other processing filters, but in our case, we were contented with these essential ones (see Fig. 2).

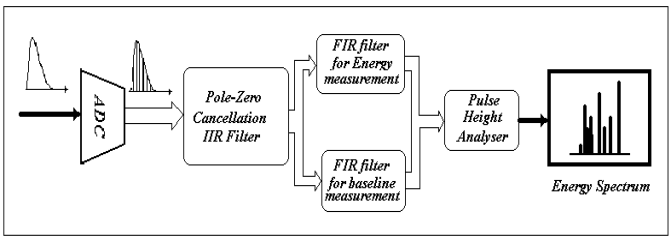


Fig. 2 Schematic representation of the three main digital filters.

The signal issued from the ADC undergoes a pole-zero cancellation operation which is necessary to cancel the long exponential decay due to the charge preamplifier. This task is performed via an IIR filter which is self calibrated so that no external intervention is asked for tuning. Baseline instabilities and fluctuations due to the thermal drifts is accomplished in an optimal manner by means of a FIR filter. The pulse's energy estimation is carried out in an optimal designed FIR filter which is managed in parallel with the baseline FIR filter. In the following, we will examine in more details the design of the three mentioned digital filters.

### *A. The Digital Filter to be Used for the "Pole-Zero/Zero-Pole" Compensation*

It is well-known, that in the majority of the gamma-ray spectroscopic setups, the slow time constant, introduced by the preamplifier part, must be eliminated before the shaping operation. Such a pretreatment is accomplished to avoid: i) *the* influence of the temporal persistence of the slow pole on the measurement of successive impulses (risk of the impulses pileup), ii) the saturation of the shaping stage and/or the ADC in the case of a high event rates coming from the detector. To conceive this pole-zero cancellation unit, we have followed the ongoing methodology.

The effects of the analog *P/Z* cancellation can be brought out by considering the transfer function of the residual couple pole/zero.

$$G(s)=\frac{1+s\tau_z}{1+s\tau_p} \qquad (01)$$

Where $\tau_p$ is the time constant introduced by the preamplifier, it is the entity which we want to cancel and $\tau_z$ is the time constant representing the zero introduced by the analog compensation circuit. To perform the digital PZ/ZP compensation, we place a digital filter just after the ADC circuit to corrects the analog error of compensation on the digitized signal, (see Fig. 3).

To design the sought digital filter, we have followed the three steps [9]:

- On-line estimation of the time constant value $\tau_P$ introduced by the preamplifier stage;
- On-line estimation of the time constant $\tau_Z$ corresponding to the zero introduced by the shaping stage.

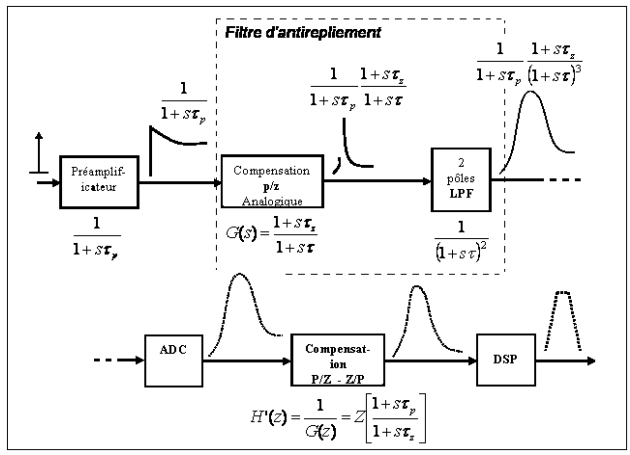


Fig. 3 Generic digital spectrometer with a partial analog P/Z and a complete Z/P digital compensation.

- Determination of the IIR digital filter taps, once the structure of the filter IIR, *H' (Z)*, was chosen.

To pass from the continuous representation to the discrete form of the filter to conceive, we have used the bilinear transformation [10].

$$s \rightarrow \frac{2}{T_s}\frac{1-z^{-1}}{1+z^{-1}} \qquad (02)$$

What gives us the expression of the digital filter which will be used to concretize the necessary P/Z compensation.

$$H'(z)=\frac{1}{G(z)}=\frac{T_s+2\tau_p}{T_s+2\tau_z}\frac{1+z^{-1}\dfrac{T_s-2\tau_p}{T_s+2\tau_p}}{1+z^{-1}\dfrac{T_s-2\tau_z}{T_s+2\tau_z}} \qquad (3)$$

The two figures below (4-5) depict the shapes of the digitized pulse before and after the "Pole-Zero/Zero-Pole" compensation

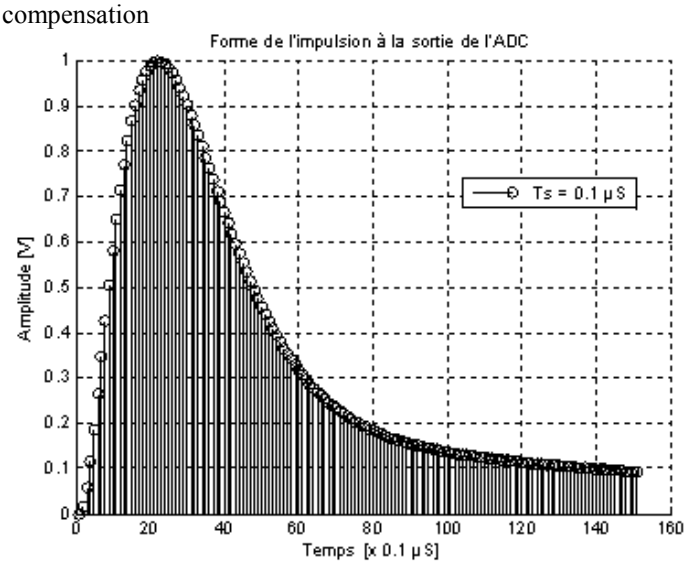


Fig. 4 Digitized nuclear pulse, before the P/Z compensation.

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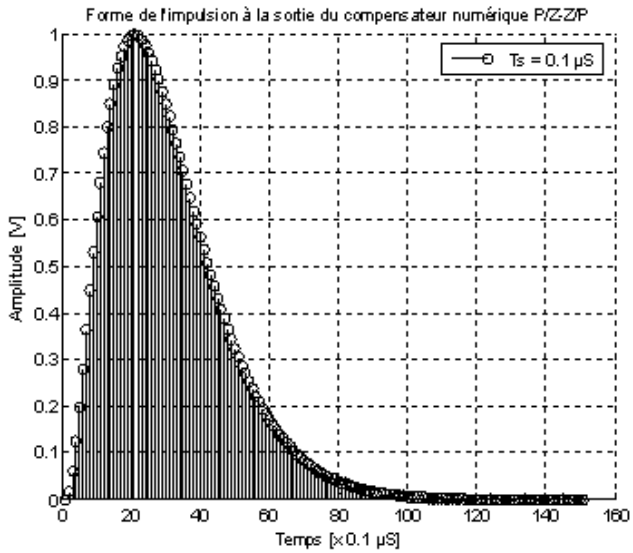


Fig. 5 Digitized nuclear pulse, after the P/Z compensation.

The comparison of the two figures confirms the effectiveness of the method undertaken to get the required filter.

### *B. The Optimal Digital Filter to be Used for the Pulse Energy Measurement*

The calculation of the optimal filter weighting functions used for the measurement of the nuclear pulse energy and the baseline estimation requires the characterization of the measuring equipment front end electronics relatively to the current noise components in the setup. Such a characterization passes inevitably by the calculation and/or the on-line measurement of the total power spectral density (psd): *N(ω)* representing the electronic noise at the input of the considered spectrometer.

*The Noise Power Spectral Density Estimation*

To determine the mathematical expression of power spectral density characterizing the electronic noise at the input of the used spectrometer "*Gamma -ray spectrometer with (HPGe) nitrogen cooled detector*", we have achieved the following tasks:

- Acquisition on PC of the temporal recordings of the signal at the output of the of the analog part (preamplifier & anti-aliasing filter) in the absence of any radioactive source. Acquisitions were made via the digital oscilloscope "Tektronix TDS220", which was connected to the PC using an RS-232D link. Several sampling rates were tested before the final choice. Fig. 6 illustrates a sample of these temporal records.
- Using these samples, we have calculated an estimate of the power spectral density through the averaged periodogram method (direct method) [10]. The task was achieved out in Matlab-R14 software.

The shape of one sample of the curves obtained at the end of this step is represented in Fig. 7.

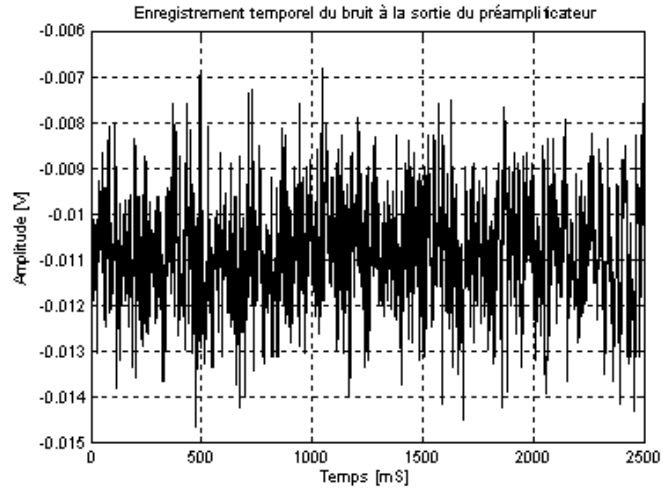


Fig. 6 Sample of the temporal noise recording at the output of the analog stage.

The power spectral density estimation thus obtained, is brought back to the input of the spectrometer by dividing its output values, point by point, by the squared transfer transfer function characterizing the analog part of the considered spectrometer. The mathematical expression of the (psd) thus obtained was then expressed in negative and positive terms of power of the frequency (Laurent series). To arrive to this result, we have used the weighted of least squares method. An example of expressions obtained (in the forms of Laurent series) is given by:

$$N(\omega) = 6.1\text{x}10^{-29} f^{0} + 4.7 \text{ x } 10^{-39} f^{2} \text{ [A}^{2}\text{/Hz]}. \quad (4)$$

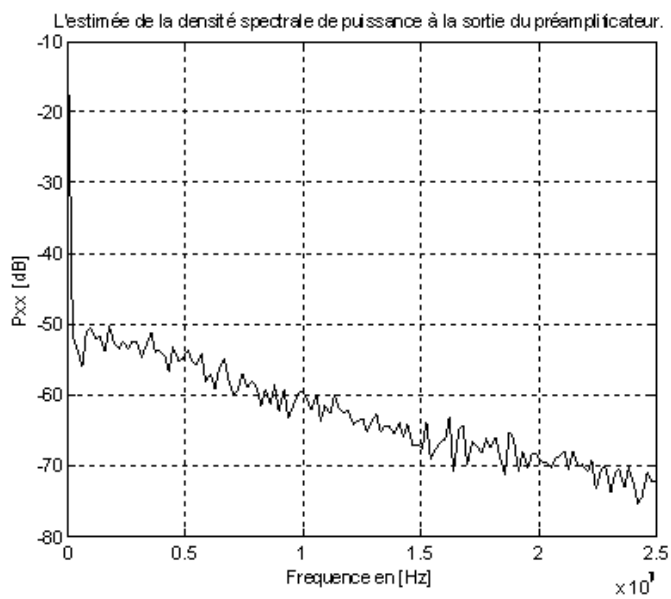


Fig. 7 Experimental noise spectral density.

*The Energy Optimal Filter Design*

This design inevitably passes by the minimization of the noise/signal ration (which is equivalent to the maximization of the signal/noise ration SNR). In our case, this ratio was calculated basing on the electric equivalent schematic of generic Gamma-ray spectroscopic measuring equipment given below [11]:

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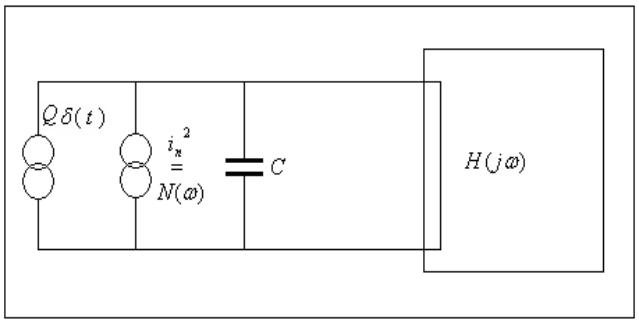


Fig. 8 Reduction of the whole noise sources to only one equivalent source (current generator) of a psd represented by $N(\omega)$.

According to this simplified schematic, the expression of the noise to signal ration (NSR) at the output of the studied setup is given by:

$$\frac{\overline{n}^2}{s^2} = \frac{b\int_{-\infty}^{+\infty} N(\varphi)W(\varphi)W^*(\varphi)d\varphi}{f_c Q^2}. \qquad (05)$$

With: $\varphi = f / f_c$, $f_c$: the noise corner frequency, $b$: the parallel component of the white noise. $N(\varphi)$ : the noise power spectral density at the input of the filter to be designed, $W(\varphi)$ : the transfer function of the considered filter and $Q$ : the electrical charge released in the detector.

To get the parameters of the optimal sought filter, we have to minimize this entity with respect to the weighting function $W(\varphi)$. Which have the same meaning as dealing with a problem of optimization under constraints (constraints are here applied to the weighting function: Limited duration, flat-topper shape, and/or null total surface). To solve such a problem, we can use many mathematical available methods in the field of functional analysis. Concerning this application, we have used the "variational methods" [11]. Which yields to the final expression of the *weighting function of the optimal sought and constrained filter?*

$$\boxed{W_{opt}(\varphi) = \frac{1}{N(\varphi)}(C_0 + C_1 e^{-i\varphi\tau_1} + C_2 e^{-i\varphi\tau_2} + ...)} \qquad (06)$$

While basing on the theoretical concepts quoted above, we use a software "under Matlab-R14" which allows us to calculate the optimal weighting function $W_{opt}(\varphi)$ (under or without temporal constraints) relating to the filter necessary for the measurement of energy released by an ionizing radiation in the used detector. The inputs to the procedure were: the shape of the useful signal and the power spectral density of the actual noise noise present at the input of the spectrometer under study. Examples of the shapes obtained after the executions of the weighting function calculation program are illustrated in Figs. 9 and 10.

### *C. The Optimal Digital Filter to be Used for the Baseline Estimation*

In an analog Gamma-ray spectroscopic setup, signals seen at the output of the amplification and shaping modules are generally overlapped on a shifted voltage of the baseline (0V).

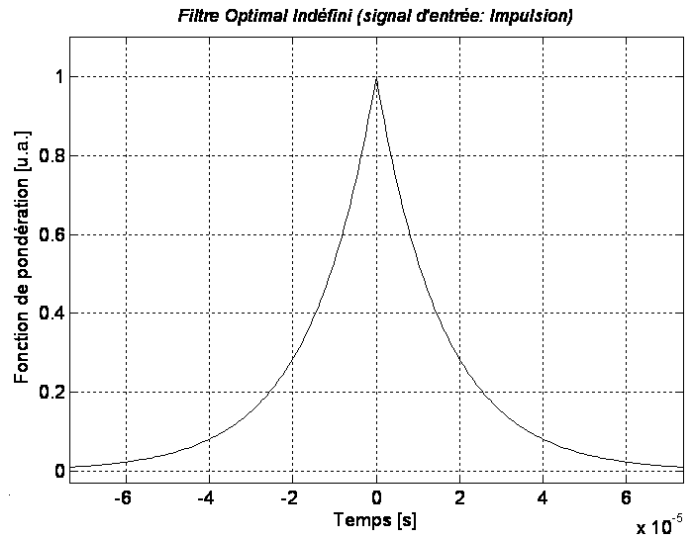


Fig. 9 Optimal weighting function characterizing the optimal filter to be used for the energy measurement (no applied constraints).

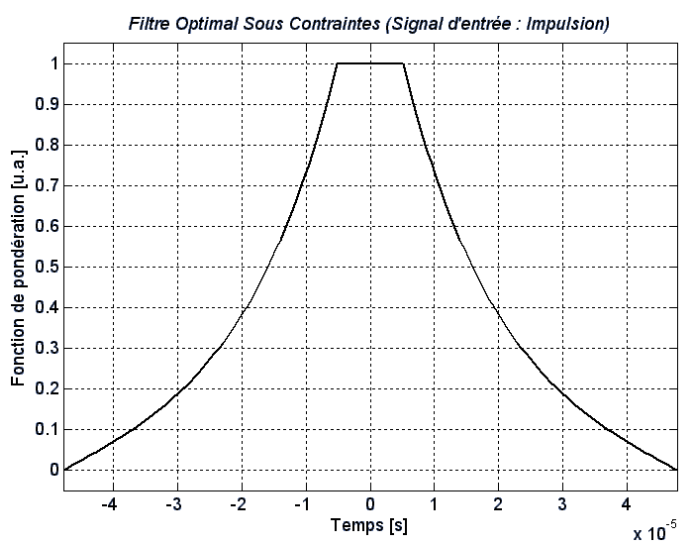


Fig. 10 Optimal weighting function characterizing the optimal filter to be used for the energy measurement (Applied constraints: Flat-top, Time limitation).

In practical situations, this shift is not usually stable, because it depends on the leakage current of the detector (*coupled DC systems with radiation rates inducing the pileup phenomenon, or AC coupled systems with a weak P/Z compensation)* [9] and the thermal drifts of the electronic components. In order to reduce the inaccuracy and the widening of spectral measurements, the amplitude of the base line is generally considered, then subtracted from the heights of the measured impulses using a judicious circuit commonly called the (*Base Line Restorer "BLR"*).

### *The Baseline Estimation Optimal Filter Design*

In the context of a shifted base line, a Gamma-ray spectrometer is electrically schematized by the figure reproduced below [12]. We have used this schematic during all

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our optimal baseline estimation filter design. The followed methodology yields to a digital filter which gives an optimized estimation of the base line level seen at the spectrometer's output.

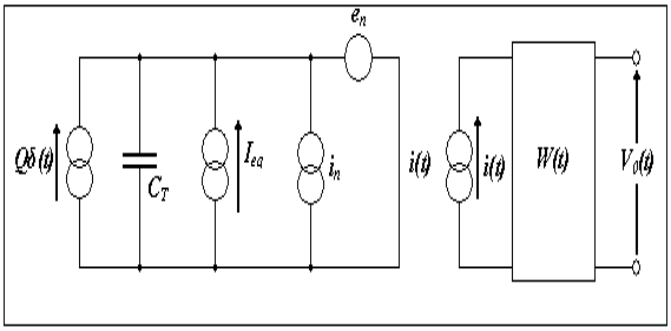


Fig. 11 Basic electrical schematic of a Gamma-ray spectrometer in the context of a shifted baseline.

While basing on this schematic and by always using the "variational methods" to maximize the signal to noise ratio at the output of the sought filter. We can easily bet the weighting function characterizing the minimum noise filter noted $W_{MN}(t)$ which has as an expression:

$$\begin{cases} W_{MN}(t) = \dfrac{1/2}{T_f - T_c \tan\alpha}\left(1 - \dfrac{\cosh(\alpha t / T_f)}{\cosh\alpha}\right) \cdot \bar{A} & pour \quad -T_f \le t \le T_f \\ W_{MN}(t) = 0 & pour \quad t \le -T_f \quad et \quad t \ge T_f \end{cases} \qquad (07)$$

A concrete example of a weighting function characterizing the optimal filter used for the evaluation of the shifted baseline is given in Fig. 12.

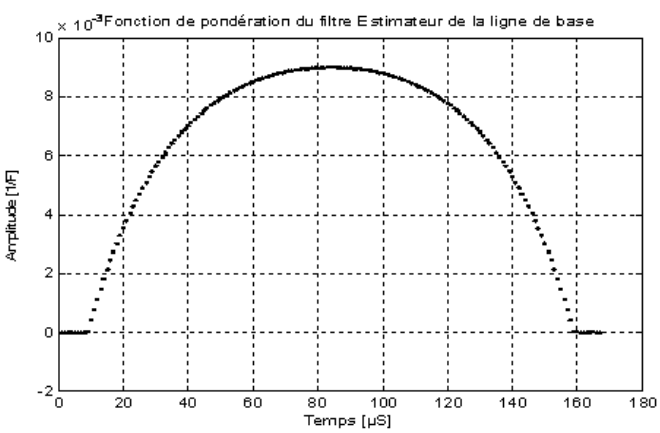


Fig. 12 Weighting function characterizing the optimal filter to be used for the evaluation of the shifted baseline.

This filter's parameters, which are obtained from the quoted above weighting function, were use to process a *DC* noised signal (see Fig. 13.) representing a shifted baseline.

As results, we get:

- Introduced value [u. a.] = *+ 2*
- Estimated value [u. a.] = *+ 1.9965*

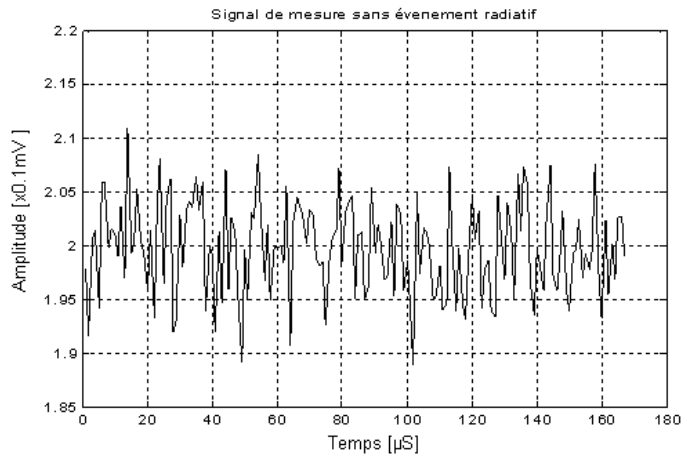


Fig. 13 Noisy baseline shifted towards the positive side.

At the end of this operation, we can notice that the results given by the designed filter is very close to that theoretically introduced.

## III. DSP Implementation of the Designed Digital Filters

DSPs (*Digital Signal Processors)* are processors especially dedicated to the digital signal processing; Digital filtering is a particular case of the processing task. The design of the digital filters to be implemented on DSP chip generally poses the following problems:

*Approximation*: Which consists in generating a transfer function which satisfying the whole specific conditions in the frequency and/or temporal domain. This part was examined in the preceding sections.

*Realization*: several types of structures are possible to fulfill a given transfer function given (direct form, cascades, parallel,…etc.). It is the part which we will develop in this section.

The whole filters designed within the framework of this work were implemented and tested separately on a DSP Board (see table I) referenced as "*TMS320C1367 DSK*" and made by Digital Spectrum [13].

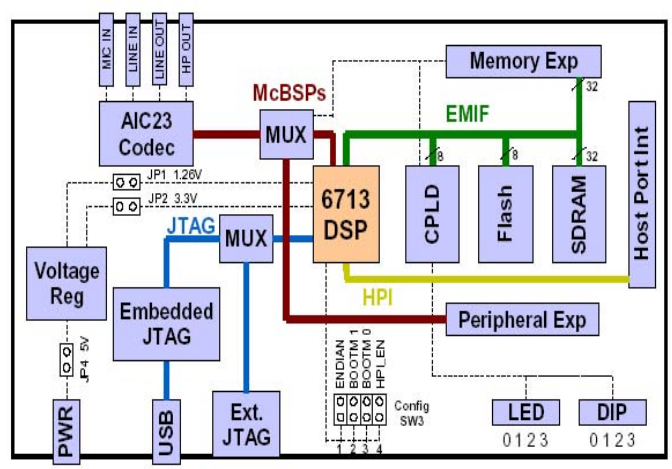


Fig. 14 The Block diagram of the DSP used for the designed digital filters implementation.

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| ***Digital Filter*** | ***Function*** |
|---|---|
| Direct Forme (First order) *IIR* Filter. | Digital P/Z Compensation Filter. |
| Direct Forme (56 coeff.: $a_0 \sim a_{55}$) *FIR* Filter. | Nuclear pulse Energy measurement digital Filter. |
| Direct Forme (167 coeff.:$a_0 \sim a_{166}$) *FIR* Filter. | Digital Baseline Restorer Filter (BLR). |

Table I. The implemented Digital Filters

## IV. CONCLUSIONS

Basing on concepts stipulating that in the high resolution nuclear spectroscopy, the charge $Q$ created in the ionizing radiations detector must be measured with a maximum of possible accuracy and precision. The subject to find the best digital filters for this charge measurement under variable experimental noise components was carefully examined along this paper. The filters obtained in the design phase could not be efficiently implemented without a beneficial contribution of the digital signal processing techniques. A tool which facilitated the implementation of the filters and this in spite of their specific characteristics, such the insertion of a flat top to the total temporal response of the measuring equipment, and whose realization was impossible very near via an analog processing tools.

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